

Methodological Note (v5)

Replication of Boucher, Rendall, Ushchev and Zenou (2024)

Toward a General Theory of Peer Effects

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June 2026

Abstract. This note documents the consolidated final code (*v5*) for the replication. It explains, with the math worked out and every symbol defined, the structural model, why estimation uses GMM, where identification comes from, the concentrated two-step estimator and the continuously updated estimator (CUE), and the full battery of diagnostics. It adds two things relative to earlier drafts: a comparison of several instrument transformations (to see which identify β worse) and the over-identification J-test. All figures in the results come from the runs with 150 schools and 30,000 students. The estimation is clean because the data come from a correctly specified synthetic DGP; this validates method and code, not an empirical claim about peer effects.

1. Goal and strategy

The paper generalizes the *linear-in-means* (LIM) peer model, where each agent is influenced by the *average* of their peers, to a CES social norm with a curvature parameter β that decides *which* peers one responds to. With no access to the real microdata (Add Health), we run a Monte Carlo: (i) generate synthetic data *from the model* with known true parameters; (ii) estimate; (iii) check recovery. If we know the truth, we can validate the estimator and the code.

1.1 Notation

2. The structural model

For each student i ,

$$\text{isolated: } y_i = p_i, \quad \text{connected: } y_i = \delta p_i + \lambda S_i(\beta), \quad (1)$$

- y_i — GPA; p_i — private component; $S_i(\beta)$ — peer norm.
- δ — weight on own private component; λ — weight on peers.
- isolated = no friends (no peer term); connected = at least one friend.

The private component is

$$p_i = c + x_i^\top \gamma + u_{s(i)} + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid x, G, u] = 0. \quad (2)$$

- c — common intercept; $x_i^\top \gamma$ — covariate part; $u_{s(i)}$ — school fixed effect.
- ε_i — mean-zero idiosyncratic error given observables.

2.1 What β does: a power mean

$$S_i(\beta) = \left(\sum_j g_{ij} y_j^\beta \right)^{1/\beta}, \quad \sum_j g_{ij} = 1. \quad (3)$$

Table 1. Symbols used throughout.

Model	
i, j	student indices (j : friends of i); $s(i)$ school of i
y_i	observed GPA (the outcome) of student i
p_i	private component: i 's grade absent peer influence
x_i, γ, c	covariates (age, female, college parent), their coefficients, intercept
$u_{s(i)}, \varepsilon_i$	school fixed effect; idiosyncratic error, $\mathbb{E}[\varepsilon_i x, G, u] = 0$
g_{ij}, G	weight i gives friend j (row-normalized); network matrix
$S_i(\beta)$	CES social norm of i 's peers
β	curvature of the norm (which peers count)
λ, δ	peer-effect intensity; weight on the private component
$\lambda_1 = \lambda + \delta - 1$	endogenous spillover (contagion)
$\lambda_2 = 1 - \delta$	conformism
Estimation	
$\hat{y}_i$	first-stage predicted GPA (own covariates + school FE)
$\hat{S}_i$	norm at $\hat{y}$: instrument for the level of the norm
$\hat{D}_i = \partial \hat{S}_i / \partial \beta$	instrument that identifies β
$Z_i, r_i(\theta)$	instrument vector; structural residual
$\theta = (\gamma, \lambda, \beta, \delta)$	parameters; $\bar{g}(\theta) = \frac{1}{n} \sum_i Z_i r_i(\theta)$ moments
W, Ω	GMM weight matrix; variance of the moments
$Q(\theta)$	GMM objective $\bar{g}^\top W \bar{g}$
$J(\beta_0), J = nQ$	Anderson–Rubin statistic; Hansen over-identification statistic
n, G_{cl}	students (30,000); schools / clusters (150)

- g_{ij} — friendship weights (sum to 1 $\Rightarrow$ weighted average); y_j — friends' grades.
- β — curvature: $\beta = 1$ mean (LIM); $\beta > 1$ over-weights high achievers; $\beta \rightarrow \infty$ the max.

LIM is the special case $\beta = 1$; the empirical question is which β fits (here the true value is 10).

2.2 Reparametrization and equilibrium

$$\lambda_1 = \lambda + \delta - 1, \quad \lambda_2 = 1 - \delta, \quad y_i = p_i + \underbrace{\lambda_1 S_i}_{\text{contagion}} + \underbrace{\lambda_2 (S_i - p_i)}_{\text{conformism}}. \quad (4)$$

Because y_i depends on peers' y_j , the school's outcomes solve a fixed point $y = T(y)$ with $T(y) = \delta p + \lambda S(y; \beta)$. The CES norm is homogeneous of degree 1, its Jacobian has row sums ≤ 1 , so T is a contraction with modulus $\leq \lambda < 1$: a unique equilibrium, computed by iteration (≈ 16 steps). At $\beta = 1$, $y = (I - \lambda G)^{-1} \delta p$ and a uniform shock is amplified by the *social multiplier* $\delta/(1 - \lambda) = 0.85/0.75 = 1.133$.

- λ_1 — pure spillover; λ_2 — pressure toward the group norm; $\lambda_1 > 0 \iff \delta + \lambda > 1$.
- $(I - \lambda G)^{-1} = I + \lambda G + \lambda^2 G^2 + \dots$ — sum of all feedback rounds.

3. The econometric problem

3.1 Reflection and endogeneity

S_i contains the friends' y_j , which depend on ε_j and, through the equilibrium feedback, on ε_i . Hence

$$\mathbb{E}[S_i \varepsilon_i] \neq 0, \quad (5)$$

so OLS of y_i on S_i is inconsistent: $\hat{\lambda}_{\text{OLS}} \xrightarrow{p} \lambda + \text{Cov}(S_i, \varepsilon_i) / \text{Var}(S_i)$. This is Manski's (1993) reflection problem plus simultaneity.

3.2 Why GMM (not OLS or MLE)

GMM (Hansen, 1982) needs only *instruments* Z_i with $\mathbb{E}[Z_i r_i(\theta_0)] = 0$. We use it because β enters *nonlinearly* (OLS cannot) and because GMM is semiparametric (no distributional assumption on ε , unlike MLE) and robust to clustering; with optimal weights it is efficient.

4. Identification and instruments

4.1 Network intransitivities and isolated students

Bramoullé, Djebbari and Fortin (2009): in a triad $A \rightarrow B \rightarrow C$ with $A \not\rightarrow C$, C 's characteristics shift A 's norm (through B) but do not enter A 's equation, so they instrument S_A . Network gaps (intransitivities) create the instruments. Separately, isolated students obey $y_i = c + x_i^\top \gamma + u + \varepsilon_i$ with no endogeneity, so they pin down γ directly (hence γ is very precise).

4.2 The baseline instruments $\hat{S}$ and $\hat{D}$

The first stage predicts $\hat{y}$ from own covariates and school effects, then builds

$$\hat{S}_i = \left(\sum_j g_{ij} \hat{y}_j^\beta \right)^{1/\beta}, \quad \hat{D}_i = \frac{\partial \hat{S}_i}{\partial \beta} = \hat{S}_i \left[\frac{1}{\beta} \frac{\sum_j g_{ij} \hat{y}_j^\beta \ln \hat{y}_j}{\sum_j g_{ij} \hat{y}_j^\beta} - \frac{1}{\beta^2} \ln \sum_j g_{ij} \hat{y}_j^\beta \right]. \quad (6)$$

- $\hat{y}_j$ — friend's predicted grade (function of exogenous x only).
- $\hat{S}_i$ — instruments the *level* of the norm $\Rightarrow$ identifies λ .
- $\hat{D}_i = \partial \hat{S}_i / \partial \beta$ — instruments the *curvature* $\Rightarrow$ identifies β . It is tiny at high β (the norm saturates), which is why β is the hard parameter.
- The baseline does *not* use friend-of-friend covariates (G^2X): the first stage uses only own covariates, so $\hat{S}, \hat{D}$ already encode friends' (distance-1) characteristics. G^2X is offered as an option (below).

The non-isolated moments stack $Z_i = (x_i, \hat{S}_i, \hat{D}_i)$ (5 moments) and the isolated $Z_i = x_i$ (3): 8 moments for 6 parameters, over-identified by 2.

4.3 Configurable instruments [v5]

v5 makes the non-isolated instrument set a switch (`ACTIVE_INSTRUMENTS`). Beyond the baseline, one can append $\hat{D}^2$, $\hat{S}^2$, the norm at $\beta = 1$, friends' covariates GX , or friends-of-friends G^2X , and `compare_instrument_sets` refits under each to show which estimate β worse (Section 7.4).

5. The estimator

5.1 Concentrated two-step GMM

(a) Within transform by (school $\times$ isolation) removes $u_{s(i)}$ and c . (b) Given (λ, β, δ) , γ is linear, solved analytically, so the search is over 3 parameters:

$$Q(\lambda, \beta, \delta) = \bar{g}(\theta)^\top W \bar{g}(\theta), \quad \hat{\theta} = \arg \min_{\theta} Q(\theta). \quad (7)$$

(c) Two steps: $W = I$ first ($\hat{\theta}_1$), then $\widehat{W} = \widehat{\Omega}^{-1}$ with $\widehat{\Omega} = \frac{1}{n} \sum_i Z_i Z_i^\top r_i(\hat{\theta}_1)^2$, re-optimized for the efficient $\hat{\theta}_2$. A global β prescan plus multistart avoid spurious optima.

- W — weight matrix; $\widehat{W} = \widehat{\Omega}^{-1}$ is the efficient choice.

5.2 CUE [v3 / v5]

The continuously updated estimator recomputes the weight matrix at *every* θ inside the objective:

$$Q_{\text{CUE}}(\theta) = \bar{g}(\theta)^\top \Omega(\theta)^{-1} \bar{g}(\theta). \quad (8)$$

- $\Omega(\theta)$ — moment variance recomputed at each θ (vs fixed in two-step).
- Motivation: with a weak instrument CUE can in principle have better finite-sample behavior. Finding (Section 7.5): in this correctly specified DGP it yields *no practical improvement* over the two-step estimator.

6. Inference

Cluster-robust SE (sandwich, school clusters, with $G_{\text{cl}}/(G_{\text{cl}} - 1)$):

$$\widehat{V} = (\widehat{G}^\top W \widehat{G})^{-1} \widehat{G}^\top W \widehat{\Lambda} W \widehat{G} (\widehat{G}^\top W \widehat{G})^{-1}, \quad \widehat{\Lambda} = \frac{G_{\text{cl}}}{G_{\text{cl}} - 1} \sum_s \left(\sum_{i \in s} Z_i r_i \right) \left(\sum_{i \in s} Z_i r_i \right)^\top. \quad (9)$$

- $\widehat{G} = \partial \bar{g} / \partial \theta^\top$ — moment Jacobian; $\widehat{\Lambda}$ — school-clustered moment variance.
- λ_1, λ_2 SEs by the delta method; clustering reflects 150 schools, not 30,000 independent students.

Anderson–Rubin (weak-IV robust): fix $\beta = \beta_0$, profile the rest, $J(\beta_0) = n Q(\beta_0) \xrightarrow{d} \chi_3^2$; the 95% set is $\{\beta_0 : J(\beta_0) \leq 7.81\}$. **Hansen J -test [v5]** (the extra moments as a test): at the efficient step $J = n Q(\hat{\theta}) \xrightarrow{d} \chi_{8-6}^2 = \chi_2^2$; a large J rejects the over-identifying restrictions. **LIM test**: compare the objective at $\beta = 1$ to the free optimum.

7. Results

7.1 Generated data

Table 2. Synthetic environment (DGP_v5).

Schools	150	Mean GPA	2.778
Students	30,000	Min / Max GPA	1.378 / 4.000
Isolated	6,080 (20.3%)	Clipped to [1, 4]	1 (0.003%)

7.2 Parameter recovery

γ, λ, δ land almost exactly on the truth; β is the least precise. Two-step and CUE coincide at this design.

Table 3. Two-step and CUE estimates (150 schools). School-cluster robust SE; t for the two-step.

Parameter	True	Two-step	CUE	Cluster SE	t
γ_{age}	0.10	0.0985	0.0985	0.0014	71.4
γ_{female}	0.24	0.2401	0.2401	0.0032	75.2
$\gamma_{\text{parent coll.}}$	0.40	0.3984	0.3984	0.0045	89.2
λ	0.25	0.2518	0.2517	0.0076	33.2
β	10.00	9.350	9.359	1.105	8.46
δ	0.85	0.8486	0.8486	0.0091	92.9
λ_1	0.10	0.1004	0.1003	0.0116	8.66
λ_2	0.15	0.1514	0.1514	0.0091	16.57
GMM objective Q	—	0.000185	0.000185		

Table 4. Instrument strength and weak-IV / over-identification / LIM tests.

Partial F (for β)	38.5	AR 95% set for β	[9.0, 10.5]
First-stage R^2 ($\hat{y}$)	0.575	$\chi^2_{0.95,3}$	7.81
$\text{corr}(S, \hat{S})$	0.701	Hansen $J = nQ$ (dof 2)	5.56 ($p = 0.062$)
mean $\hat{D}$ (levels)	0.0041	LIM obj. vs free	0.0071 vs 0.00019

7.3 First stage, instruments and tests

The AR set contains the true $\beta = 10$. The J -test does not reject the model (close, $p = 0.06$, single draw). The LIM ($\beta = 1$) is rejected: the distance statistic $n(Q_{\text{LIM}} - Q_{\text{free}}) \approx 209$ on χ^2_1 gives $p \approx 0$ — the norm is not the simple mean.

7.4 Instrument transformations

Table 5. Refitting under different instrument sets (illustrative, 30 schools, true $\beta = 10$). Read: β far from 10, huge SE, or tiny $F \Rightarrow$ worse.

Instrument set	$\hat{\beta}$	SE(β)	F
baseline $[X, \hat{S}, \hat{D}]$	8.85	2.22	8.9
+ $\hat{D}^2$ (the trap)	8.10	2.08	8.7
+ $\hat{S}^2$	10.73	3.10	8.9
+ $\hat{S}(\beta=1)$	8.39	2.04	8.8
+ GX (friends)	8.44	2.04	8.8
+ G^2X (friends of friends)	9.38	2.15	9.0
$\hat{S}$ only (no $\hat{D}$)	2.00	57.2	0.5

The qualitative ranking is robust: dropping $\hat{D}$ leaves β *unidentified* (SE = 57), confirming $\hat{D}$ carries the curvature signal; extra derivative-like moments ($\hat{D}^2$, $\hat{S}^2$) do not help and worsen the fit; friend / friend-of-friend covariates are comparable to the baseline.

Table 6. Monte Carlo, two-step vs CUE, true $\beta = 10$, two designs.

Design	Estimator	mean $\hat{\beta}$	median	RMSE	coverage
30 schools (21 reps)	Two-step	10.50	9.73	2.76	0.95
	CUE	10.55	9.73	2.78	0.95
150 schools (8 reps)	Two-step	9.91	9.82	0.54	1.00
	CUE	9.92	9.84	0.53	1.00

7.5 Two-step vs CUE: no practical improvement

The two estimators are statistically indistinguishable at both designs. At 150 schools (the real design) they differ by less than 0.05 in 100% of draws (mean $|\hat{\beta}_{\text{CUE}} - \hat{\beta}_{2s}| = 0.013$), with identical RMSE, SE and coverage; at 30 schools their medians, RMSE and tails coincide. An earlier 50-draw run had appeared to favour CUE on RMSE, but that gap was driven by a single two-step blow-up in one seed set and does not survive re-sampling. **Conclusion:** CUE recomputes $\Omega(\theta)^{-1}$ at every θ at extra computational cost and delivers *no* improvement here; we keep it as a documented option and report the two-step estimator (with the Anderson–Rubin set for β) as primary.

7.6 Monte Carlo and identification frontier

Table 7. Left: bias / 95% coverage (36 reps, 30 schools). Right: β frontier.

Param.	Bias	Cov.	Schools	SE(β)	F
λ	−0.004	0.94	20	3.14	5.9
β	+0.351	0.97	40	2.70	12.0
δ	+0.001	0.97	80	1.78	23.0
λ_1	−0.003	0.94	150	1.09	37.1
λ_2	−0.001	0.97	300	0.85	78.9

SE(β) falls and F rises with the number of schools: more (and denser) network data is the lever for β .

8. Why the estimation comes out clean

Plainly, the experiment is built to be clean, not because peer effects are easy to measure:

1. **Zero misspecification:** data come from exactly the estimated model ($Q \approx 0$).
2. **Instruments valid by construction:** network and errors independent; no homophily.
3. **No measurement error,** large n , the only confounder (school) swept exactly, and isolated students pin down γ .

The only genuine difficulty is β : at high β the CES norm saturates and $\hat{D} \rightarrow 0$. This validates **method and code**; with real data it would be messier.

9. Conclusion

The pipeline is: a CES peer model with a fixed-point equilibrium; endogeneity (reflection) that rules out OLS; identification from network intransitivities plus the isolated subsample; concentrated two-step GMM and CUE with γ profiled out and cluster-robust SE; and a battery of diagnostics (recovery, Anderson–Rubin, Hansen J , LIM, instrument-transformation comparison, two-step-vs-CUE Monte Carlo, identification frontier). All eight parameters are recovered with near-nominal coverage; β is consistent but weakly identified at high values, a fact the diagnostics quantify. The consolidated code lives in the `*_v5` modules; `python run_all_v5.py` reproduces everything in one pass.